РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №8

дисциплина: ***Сетевые технологии***

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Группа: НФИбд-02-23

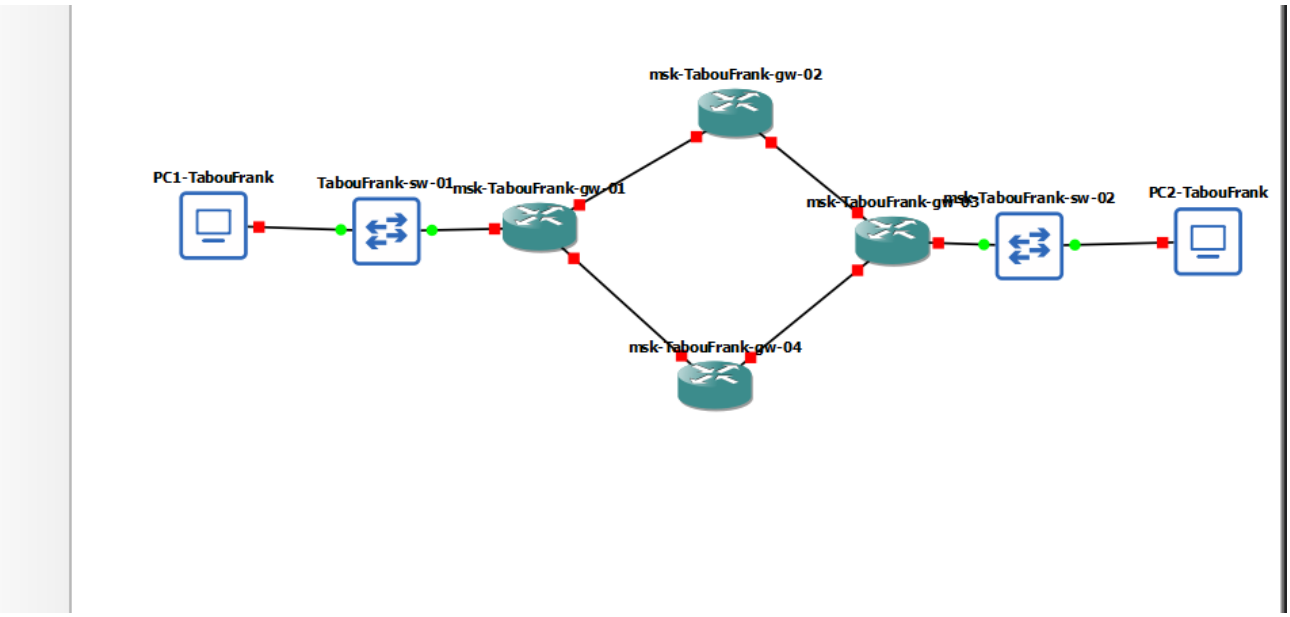
МОСКВА

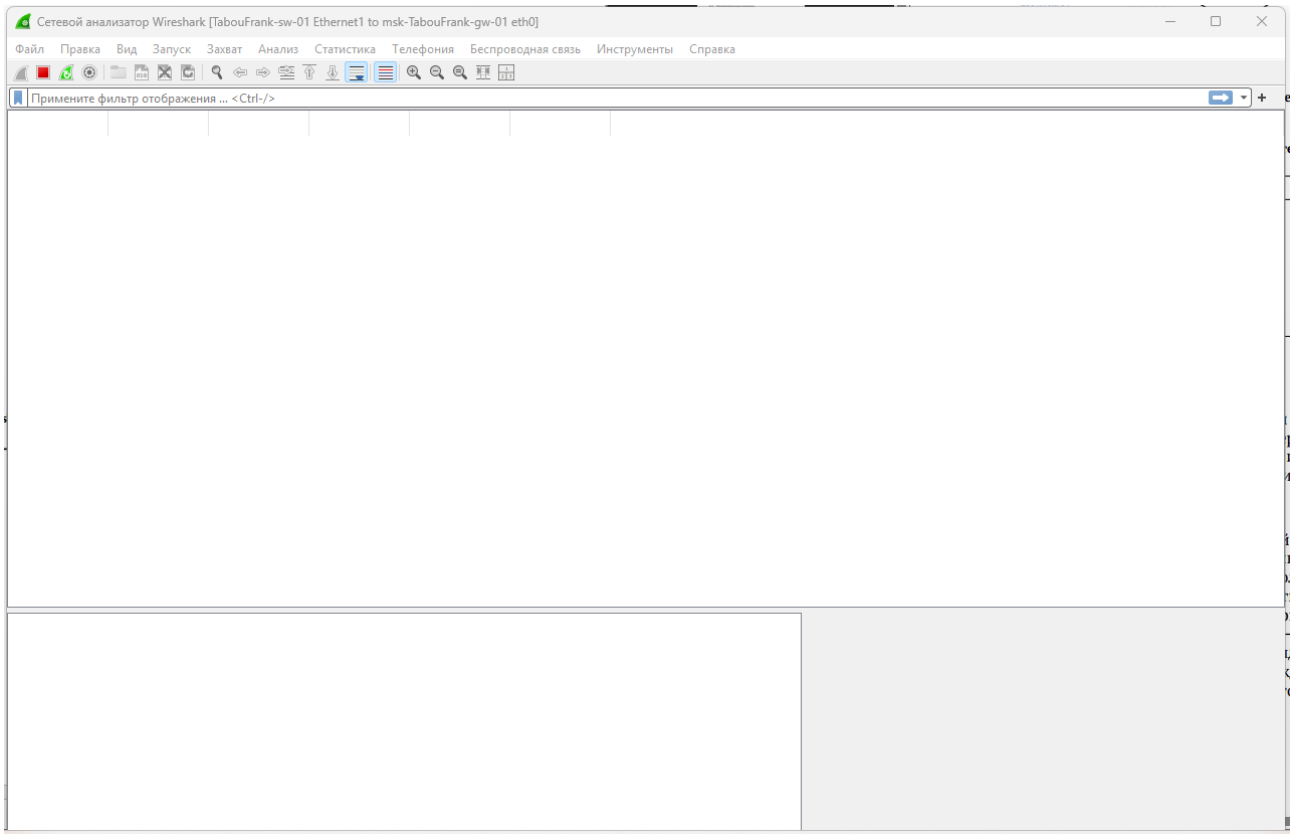
2025 г.

Цель работы:

Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

Выполнение работы:





PC1-TabouFrank - PuTTY

```
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.0.10.10/24
GATEWAY     : 10.0.10.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10016
RHOST:PORT  : 127.0.0.1:10017
MTU         : 1500

VPCS>
```

PC2-TabouFrank - PuTTY

```
VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> shop ip
Bad command: "shop ip". Use ? for help.

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.0.11.10/24
GATEWAY     : 10.0.11.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10036
RHOST:PORT  : 127.0.0.1:10037
MTU         : 1500

VPCS>
```

```
 * Setting keymap ... [ ok ]
 * Starting networking ... *   lo ... [ ok ]
 * Starting busybox syslog ... [ ok ]
 * Seeding 256 bits and crediting
 * Saving 256 bits of creditable seed for next boot
 * Starting busybox acpid ... [ ok ]
 * Starting busybox crond ... [ ok ]
Started watchfrr
 * Starting sshd ... [ ok ]


Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-frank-gw-01
msk-frank-gw-01(config)# interface eth0
msk-frank-gw-01(config-if)# ip address 10.0.10.1/24
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth1
msk-frank-gw-01(config-if)# ip address 10.0.1.1/24
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)#
```

```
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-frank-gw-01#
```

msk-frank-gw-02 - PuTTY

```
frr(config)# hostname msk-frank-gw-02
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# ip address 10.0.2.1/24
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)#
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-02
service integrated-vtysh-config
!
interface eth0
```

```
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth2
msk-frank-gw-03(config-if)# ip address 10.0.3.1/24
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# exit
msk-frank-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-03
service integrated-vtysh-config
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
interface eth0
 ip address 10.0.11.1/24
```

msk-frank-gw-04 - PuTTY

```
Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname-frank-gw-04
% Unknown command: hostname-frank-gw-04
frr(config)# hostname msk-frank-gw-04
msk-frank-gw-04(config)# interface eth0
msk-frank-gw-04(config-if)# ip address 10.0.3.2/24
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ip address 10.0.4.1/24
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-frank-gw-04#
```

PC1-TabouFrank - PuTTY

```
MAC          : 00:50:79:66:68:00
LPORT        : 10016
RHOST:PORT   : 127.0.0.1:10017
MTU          : 1500

VPCS> ip 2001:10::a/64
PC1 : 2001:10::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME         : VPCS[1]
IP/MASK      : 10.0.10.10/24
GATEWAY      : 10.0.10.1
DNS          :
MAC          : 00:50:79:66:68:00
LPORT        : 10016
RHOST:PORT   : 127.0.0.1:10017
MTU          : 1500

VPCS>
```

PC2-TabouFrank - PuTTY

```
LPORT        : 10036
RHOST:PORT   : 127.0.0.1:10037
MTU          : 1500

VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ipv6

NAME               : VPCS[1]
LINK-LOCAL SCOPE   : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE       : 2001:11::a/64
DNS                :
ROUTER LINK-LAYER  :
MAC                : 00:50:79:66:68:01
LPORT              : 10036
RHOST:PORT         : 127.0.0.1:10037
MTU:               : 1500

VPCS>
```

msk-frank-gw-01 - PuTTY

```
 ip address 10.0.4.2/24
exit
!
end
msk-frank-gw-01# configure terminal
msk-frank-gw-01(config)# ipv6 forwarding
msk-frank-gw-01(config)# interface eth0
msk-frank-gw-01(config-if)# ipv6
% Command incomplete: ipv6
msk-frank-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-frank-gw-01(config-if)# no ipv6 nd suppress-ra
msk-frank-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth1
msk-frank-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth2
msk-frank-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# exit
msk-frank-gw-01#
```

msk-frank-gw-02 - PuTTY

```
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.2.1/24
exit
!
end
msk-frank-gw-02# configure terminal
msk-frank-gw-02(config)# ipv6 forwarding
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# interface eth1
msk-frank-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

msk-frank-gw-02 - PuTTY

```
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.2.1/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ipv6 address 2001:2::1/64
exit
!
end
msk-frank-gw-02#
```

msk-frank-gw-03 - PuTTY

```
service integrated-vtysh-config
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
interface eth0
 ip address 10.0.11.1/24
exit
!
end
msk-frank-gw-03# configure interface
% Unknown command: configure interface
msk-frank-gw-03# configure terminal
msk-frank-gw-03(config)# ipv6 forwarding
msk-frank-gw-03(config)# interface eth0
msk-frank-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-frank-gw-03(config-if)# no ivp6 nd suppress-ra
% Unknown command: no ivp6 nd suppress-ra
msk-frank-gw-03(config-if)# no ipv6 nd suppress-ra
msk-frank-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth1
msk-frank-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)#
```

msk-frank-gw-03 - PuTTY

```
interface etho
 ip address 10.0.11.1/24
exit
!
end
msk-frank-gw-03# configure interface
% Unknown command: configure interface
msk-frank-gw-03# configure terminal
msk-frank-gw-03(config)# ipv6 forwarding
msk-frank-gw-03(config)# interface eth0
msk-frank-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-frank-gw-03(config-if)# no ivp6 nd suppress-ra
% Unknown command: no ivp6 nd suppress-ra
msk-frank-gw-03(config-if)# no ipv6 nd suppress-ra
msk-frank-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth1
msk-frank-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth2
msk-frank-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# exit
msk-frank-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03#
```

msk-frank-gw-03 - PuTTY

```
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-03
service integrated-vtysh-config
!
interface eth0
 ipv6 address 2001:11::1/64
 ipv6 nd prefix 2001:11::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.2.2/24
 ipv6 address 2001:2::2/64
exit
!
interface eth2
 ip address 10.0.3.1/24
 ipv6 address 2001:3::1/64
exit
!
interface eth0
lines 1-23
```

msk-frank-gw-04 - PuTTY

```
 ip address 10.0.4.1/24
exit
!
end
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# ipv6 forwarding
msk-frank-gw-04(config)# ipv6 address 2001:3::2/64
% Unknown command: ipv6 address 2001:3::2/64
msk-frank-gw-04(config)# interface eth0
msk-frank-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
exit
!
end
msk-frank-gw-04#
```

msk-frank-gw-01 - PuTTY

```
 ip address 10.0.1.1/24
 ipv6 address 2001:1::1/64
exit
!
interface eth2
 ip address 10.0.4.2/24
 ipv6 address 2001:4::2/64
exit
!
end
msk-frank-gw-01# configure terminal
msk-frank-gw-01(config)# router rip
msk-frank-gw-01(config-router)# version 2
msk-frank-gw-01(config-router)# network eth0
msk-frank-gw-01(config-router)# network eth1
msk-frank-gw-01(config-router)# network eth2
msk-frank-gw-01(config-router)# exit
msk-frank-gw-01(config)# exit
msk-frank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-01#
```

msk-frank-gw-02 - PuTTY

```
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ipv6 address 2001:2::1/64
exit
!
end
msk-frank-gw-02# configure terminal
msk-frank-gw-02(config)#
msk-frank-gw-02(config)# router rip
msk-frank-gw-02(config-router)# version 2
msk-frank-gw-02(config-router)# network eth0
msk-frank-gw-02(config-router)# network eth1
msk-frank-gw-02(config-router)# network eth2exit
msk-frank-gw-02(config-router)# network eth2
msk-frank-gw-02(config-router)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

msk-frank-gw-04 - PuTTY

```
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
exit
!
end
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# router rip
msk-frank-gw-04(config-router)# version 2
msk-frank-gw-04(config-router)# network eth0
msk-frank-gw-04(config-router)# network eth1
msk-frank-gw-04(config-router)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04#
```

```
msk-frank-gw-04# show ip router rip
% Unknown command: show ip router rip
msk-frank-gw-04#  show ip router rip
% Unknown command:  show ip router rip
msk-frank-gw-04#  show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:03:14
R>* 10.0.2.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:03:17
R>* 10.0.10.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:03:14
msk-frank-gw-04#  show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

     Network            Next Hop         Metric From            Tag Time
R(n) 10.0.1.0/24        10.0.4.2              2 10.0.4.2          0 02:47
R(n) 10.0.2.0/24        10.0.3.1              2 10.0.3.1          0 02:49
C(i) 10.0.3.0/24        0.0.0.0               1 self              0
C(i) 10.0.4.0/24        0.0.0.0               1 self              0
R(n) 10.0.10.0/24       10.0.4.2              2 10.0.4.2          0 02:47
msk-frank-gw-04#
```

```
msk-frank-gw-04#  show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 12 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface        Send  Recv   Key-chain
    eth0             2     2
    eth1             2     2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway          BadPackets BadRoutes  Distance Last Update
    10.0.3.1                 0         0       120   00:00:08
    10.0.4.2                 0         0       120   00:00:08
  Distance: (default is 120)
msk-frank-gw-04#
```

```
msk-frank-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/3] via 10.0.4.1, eth2, weight 1, 00:04:38
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:04:38
msk-frank-gw-01# show ip  rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

     Network            Next Hop         Metric From            Tag Time
C(i) 10.0.1.0/24        0.0.0.0               1 self              0
R(n) 10.0.2.0/24        10.0.4.1              3 10.0.4.1          0 02:42
R(n) 10.0.3.0/24        10.0.4.1              2 10.0.4.1          0 02:42
C(i) 10.0.4.0/24        0.0.0.0               1 self              0
C(i) 10.0.10.0/24       0.0.0.0               1 self              0
msk-frank-gw-01# show ip  rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 27 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface        Send  Recv   Key-chain
    eth0             2     2
    eth1             2     2
    eth2             2     2
  Routing for Networks:
    eth0
```

PC1-TabouFrank - PuTTY

```
MTU          : 1500

VPCS> ping 10.0.11.10

*10.0.10.1 icmp_seq=1 ttl=64 time=1.073 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=2 ttl=64 time=1.484 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=3 ttl=64 time=2.758 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=4 ttl=64 time=3.314 ms (ICMP type:3, code:0, Destination net
work unreachable)
10.0.11.10 icmp_seq=5 timeout

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1   *10.0.10.1   1.450 ms (ICMP type:3, code:0, Destination network unreachable
)  *
 2      *  *  *
 3      *  *  *
 4      *  *  *
 5      *  *  *
 6      *  *  *
```

```
msk-frank-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

     Network            Next Hop         Metric From            Tag Time
C(i) 10.0.1.0/24        0.0.0.0               1 self              0
R(n) 10.0.2.0/24        10.0.4.1              3 10.0.4.1          0 02:35
R(n) 10.0.3.0/24        10.0.4.1              2 10.0.4.1          0 02:35
C(i) 10.0.4.0/24        0.0.0.0               1 self              0
C(i) 10.0.10.0/24       0.0.0.0               1 self              0
msk-frank-gw-01#
```

1. Создай класс `Vector(x, y)`.
Реализуй:

- `__add__` – сложение векторов,
- `__sub__` – вычитание,
- `__mul__` – умножение на число,
- `__eq__` – сравнение,
- `length()` – длина вектора,
- `__str__()` – красивый вывод: `"Vector(x, y)"`.

2. Создай класс `Matrix`, принимающий двумерный список чисел.
Реализуй:

- `__add__` – сложение матриц,
- `__mul__` – умножение на число,
- `transpose()` – транспонирование,
- `__str__()` – вывод матрицы построчно.

Проверь на примере 2×2 матриц.

msk-frank-gw-01 - PuTTY

```
 * Saving 256 bits of creditable seed for next boot
 * Starting busybox acpid ... [ ok ]
 * Starting busybox crond ... [ ok ]
Started watchfrr
 * Starting sshd ... [ ok ]


Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# show ip rip
% Unknown command: show ip rip
frr(config)# router ripng
frr(config-router)# network eth0
frr(config-router)# network eth1
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr#
```

msk-frank-gw-02 - PuTTY

```
Started watchfrr
 * Starting sshd ... [ ok ]


Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure interface
% Unknown command: configure interface
frr# configure terminal
frr(config)# interface eth0
frr(config-if)# shutdown
frr(config-if)# router ripng
frr(config-router)# network eth0
frr(config-router)# network eth1
frr(config-router)# network eth2
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr#
```

msk-frank-gw-03 - PuTTY

```
 * Starting busybox acpid ... [ ok ]
 * Starting busybox crond ... [ ok ]
Started watchfrr
 * Starting sshd ... [ ok ]


Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# cofigure terminal
% Unknown command: cofigure terminal
frr# configure terminal
frr(config)# router ripng
frr(config-router)# network eth0
frr(config-router)# network eth1
frr(config-router)# network eth2
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr#
```

msk-frank-gw-04 - PuTTY

```
 * Saving 256 bits of creditable seed for next boot
 * Starting busybox acpid ... [ ok ]
 * Starting busybox crond ... [ ok ]
Started watchfrr
 * Starting sshd ... [ ok ]


Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# router ripng
frr(config-router)# network eth0
frr(config-router)# network eth1
frr(config-router)# exit
frr(config)# exit
frr# write memoey
% Unknown command: write memoey
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr#
```

*Standard input [msk-frank-gw-03 eth0 to msk-TabouFrank-sw-02 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
8	4203.576917	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
9	4804.241934	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
10	5405.085094	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
11	6005.851192	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
12	6606.339404	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
13	7208.098765	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
14	7807.759946	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
15	8408.660375	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
16	9009.094382	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
17	9609.884208	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
18	10213.137096	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
19	10812.537844	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
20	11412.960835	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
21	12013.266510	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
22	12613.799163	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
23	13214.308867	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
24	13814.715193	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
25	14415.102113	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
26	15015.717395	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
27	15616.353197	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
28	16217.002846	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
29	16817.714618	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
30	17418.478152	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
31	18019.082684	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
32	18619.710044	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
33	19220.346729	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00
34	19821.088089	fe80::e50:bdff:fe60…	ff02::1	ICMPv6	110	Router Advertisement from 0c:50:bd:60:00:00

```
> Frame 1: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface -, id 0
> Ethernet II, Src: 0c:50:bd:60:00:00 (0c:50:bd:60:00:00), Dst: IPv6mcast_01 (33:33:00:00:00:01)
> Internet Protocol Version 6, Src: fe80::e50:bdff:fe60:0, Dst: ff02::1
> Internet Control Message Protocol v6
```

```
0000  33 33 00 00 00 01 0c 50  bd 60 00 00 86 dd 60 0f   33·····P ·
0010  29 37 00 38 3a ff fe 80  00 00 00 00 00 00 0e 50   )7·8:··· ·
0020  bd ff fe 60 00 00 ff 02  00 00 00 00 00 00 00 00   ···`···· ·
0030  00 00 00 00 00 01 86 00  74 16 40 00 07 08 00 00   ····· t
0040  00 00 00 00 00 00 03 04  40 c0 00 27 8d 00 00 09   ····· @
0050  3a 80 00 00 00 00 20 01  00 11 00 00 00 00 00 00   :····· ·
0060  00 00 00 00 00 00 01 01  0c 50 bd 60 00 00         ····· ·
```

wireshark_Standard input83C5L3.pcapng | Пакеты: 34 · Потеряно: 0 (0.0%) | Профиль: Default

Захват из Standard input [TabouFrank-sw-01 Ethernet1 to msk-frank-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.10.1	224.0.0.9	RIPv2	126	Response
2	22.820365	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
3	34.006925	10.0.10.1	224.0.0.9	RIPv2	126	Response
4	67.011417	10.0.10.1	224.0.0.9	RIPv2	126	Response
5	67.824948	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
6	101.014537	10.0.10.1	224.0.0.9	RIPv2	126	Response
7	109.826729	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
8	131.017360	10.0.10.1	224.0.0.9	RIPv2	126	Response
9	133.570590	fe80::e2e:91ff:fe2f…	ff02::1	ICMPv6	110	Router Advertisement from 0c:2e:91:2f:00:00
10	151.830745	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
11	155.019884	10.0.10.1	224.0.0.9	RIPv2	126	Response
12	167.836251	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
13	189.025037	10.0.10.1	224.0.0.9	RIPv2	126	Response
14	209.837058	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
15	225.026118	10.0.10.1	224.0.0.9	RIPv2	126	Response
16	245.844109	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1

```
> Frame 1: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0
> Ethernet II, Src: 0c:2e:91:2f:00:00 (0c:2e:91:2f:00:00), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
> Internet Protocol Version 4, Src: 10.0.10.1, Dst: 224.0.0.9
> User Datagram Protocol, Src Port: 520, Dst Port: 520
> Routing Information Protocol

0000  01 00 5e 00 00 09 0c 2e  91 2f 00 00 08 00 45 c0   ··^·····  ·/····E·
0010  00 70 ba 10 40 00 01 11  ca a2 0a 00 0a 01 e0 00   ·p··@···  ····· ···
0020  00 09 02 08 02 08 00 5c  d7 06 02 02 00 00 00 02   ·····\··  ····· ···
0030  00 00 0a 00 01 00 ff ff  ff 00 00 00 00 00 00 00   ····· ··  ····· ···
0040  00 01 00 02 00 00 0a 00  02 00 ff ff ff 00 00 00   ····· ··  ····· ···
0050  00 00 00 00 00 03 00 02  00 00 0a 00 03 00 ff ff   ····· ··  ····· ···
0060  ff 00 00 00 00 00 00 00  00 02 00 02 00 00 0a 00   ····· ··  ····· ···
0070  04 00 ff ff ff 00 00 00  00 00 00 00 00 01         ····· ··  ····· ···
```

msk-frank-gw-01 - PuTTY

```
frr(config-router)# network eth1
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr# configure terminal
frr(config)# hostname msk-frank-gw-01
msk-frank-gw-01(config)# show ipv6 ripng
% Unknown command: show ipv6 ripng
msk-frank-gw-01(config)# router ospf
msk-frank-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-frank-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-frank-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-frank-gw-01(config-router)# exit
msk-frank-gw-01(config)# exit
msk-frank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-01#
```

msk-frank-gw-02 - PuTTY

```
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr# cofigure terminal
% Unknown command: cofigure terminal
frr# configure terminal
frr(config)# hostname msk-frank-gw-02
msk-frank-gw-02(config)# configure terminal
% Unknown command: configure terminal
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# router ospf
msk-frank-gw-02(config-router)# network10.0.1.0/24 area 0.0.0.0
% Unknown command: network10.0.1.0/24 area 0.0.0.0
msk-frank-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-frank-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-frank-gw-02(config-router)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

msk-frank-gw-03 - PuTTY

```
frr(config)# router ripng
frr(config-router)# network eth0
frr(config-router)# network eth1
frr(config-router)# network eth2
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr# configure terminal
frr(config)# hostname msk-frank-gw-03
msk-frank-gw-03(config)# router ospf
msk-frank-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-frank-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-frank-gw-03(config-router)# exit
msk-frank-gw-03(config)# exit
msk-frank-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03#
```

msk-frank-gw-04 - PuTTY

```
frr(config-router)# exit
frr(config)# exit
frr# write memoey
% Unknown command: write memoey
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr# cofigure terminal
% Unknown command: cofigure terminal
frr# configure terminal
frr(config)# hostname msk-frank-gw-04
msk-frank-gw-04(config)# router ospf
msk-frank-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-frank-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-frank-gw-04(config-router)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04#
```

Захват из Standard input [TabouFrank-sw-01 Ethernet1 to msk-frank-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
137	930.682182	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422786 TSecr=0 WS=2
138	931.682229	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422787 TSecr=0 WS=2
139	932.682397	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422788 TSecr=0 WS=2
140	933.560457	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
141	933.682987	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422789 TSecr=0 WS=2
142	933.747833	0c:2e:91:2f:00:00	Private_66:68:00	ARP	60	Who has 10.0.10.10? Tell 10.0.10.1
143	933.748224	Private_66:68:00	0c:2e:91:2f:00:00	ARP	60	10.0.10.10 is at 00:50:79:66:68:00
144	934.684139	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422790 TSecr=0 WS=2
145	935.684067	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422791 TSecr=0 WS=2
146	936.684747	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422792 TSecr=0 WS=2
147	937.685902	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422793 TSecr=0 WS=2
148	938.686957	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422794 TSecr=0 WS=2
149	939.687504	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422795 TSecr=0 WS=2
150	940.687383	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422796 TSecr=0 WS=2
151	941.687941	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422797 TSecr=0 WS=2
152	942.688823	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422798 TSecr=0 WS=2
153	943.560778	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
154	943.689516	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422799 TSecr=0 WS=2
155	943.923680	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
156	944.690720	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422800 TSecr=0 WS=2
157	945.691550	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422801 TSecr=0 WS=2
158	946.692577	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422802 TSecr=0 WS=2

```
Frame 1: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0
    Section number: 1
  > Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Mar 13, 2026 20:10:55.995775000 RTZ 2 (зима)
    UTC Arrival Time: Mar 13, 2026 17:10:55.995775000 UTC
    Epoch Arrival Time: 1773421855.995775000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000000000 seconds]
    [Time delta from previous displayed frame: 0.000000000 seconds]
    [Time since reference or first frame: 0.000000000 seconds]
    Frame Number: 1
    Frame Length: 126 bytes (1008 bits)
    Capture Length: 126 bytes (1008 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
```

```
0000  01 00 5e 00 00 09 0c 2e  91 2f 00 00 08 00 45 c0   ··^·····  ·/····E·
0010  00 70 ba 10 40 00 01 11  ca a2 0a 00 0a 01 e0 00   ·p··@···  ********
0020  00 09 02 08 02 08 00 5c  d7 06 02 02 00 00 00 02   ·******\  ********
0030  00 00 0a 00 01 00 ff ff  ff 00 00 00 00 00 00 00   ********  ********
0040  00 01 00 02 00 00 0a 00  02 00 ff ff ff 00 00 00   ********  ********
0050  00 00 00 00 00 03 00 02  00 00 0a 00 03 00 ff ff   ********  ********
0060  ff 00 00 00 00 00 00 00  00 02 00 02 00 00 0a 00   ********  ********
0070  04 00 ff ff ff 00 00 00  00 00 00 00 00 01         ********  ******
```

```
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-01# show ip ospf neighbor

Neighbor ID     Pri State           Up Time         Dead Time Address         In
terface                         RXmtL RqstL DBsmL
10.0.4.1          1 Full/Backup     19m00s            37.246s 10.0.4.1        et
h2:10.0.4.2                         0     0     0

msk-frank-gw-01# show ip ospf route
============ OSPF network routing table ============
N    10.0.1.0/24           [100] area: 0.0.0.0
                           directly attached to eth1
N    10.0.3.0/24           [200] area: 0.0.0.0
                           via 10.0.4.1, eth2
N    10.0.4.0/24           [100] area: 0.0.0.0
                           directly attached to eth2
N    10.0.10.0/24          [100] area: 0.0.0.0
                           directly attached to eth0

============ OSPF router routing table =============

============ OSPF external routing table ===========


msk-frank-gw-01#
```

msk-frank-gw-01 - PuTTY

```
                            directly attached to eth2
N    10.0.10.0/24           [100] area: 0.0.0.0
                            directly attached to eth0

============ OSPF router routing table =============

============ OSPF external routing table ===========


msk-frank-gw-01# show ip ospf neighbor

Neighbor ID     Pri State           Up Time         Dead Time Address         Interface
          RXmtL RqstL DBsmL
10.0.4.1          1 Full/Backup     21m39s            38.604s 10.0.4.1        eth2:10.0.4.2
              0     0     0

msk-frank-gw-01# show ip ospf route
============ OSPF network routing table ============
N    10.0.1.0/24            [100] area: 0.0.0.0
                            directly attached to eth1
N    10.0.3.0/24            [200] area: 0.0.0.0
                            via 10.0.4.1, eth2
N    10.0.4.0/24            [100] area: 0.0.0.0
                            directly attached to eth2
N    10.0.10.0/24           [100] area: 0.0.0.0
                            directly attached to eth0

============ OSPF router routing table =============

============ OSPF external routing table ===========


msk-frank-gw-01#
```

```
msk-frank-gw-01# cofigure terminal
% Unknown command: cofigure terminal
msk-frank-gw-01# configure terminal
msk-frank-gw-01(config)# router ospf6
msk-frank-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-frank-gw-01(config-ospf6)# exit
msk-frank-gw-01(config)# interface eth0
msk-frank-gw-01(config-if)# ipv6 ospf6 area 0
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth1
msk-frank-gw-01(config-if)# ipv6 ospf6 area 0
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth2
msk-frank-gw-01(config-if)# ipv6 ospf6 area 0
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# exit
msk-frank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-01#
```

msk-frank-gw-02 - PuTTY

```
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02# configure terminal
msk-frank-gw-02(config)# interface eht0
msk-frank-gw-02(config-if)# shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# interface eht0
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# cofiguration terminal
% Unknown command: cofiguration terminal
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# router ospf6
msk-frank-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-frank-gw-02(config-ospf6)# exit
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# ipv6 ospf6 area 0
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

msk-frank-gw-03 - PuTTY

```
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03# configure terminal
msk-frank-gw-03(config)# router ospf6
msk-frank-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-frank-gw-03(config-ospf6)# exit
msk-frank-gw-03(config)# interface eth0
msk-frank-gw-03(config-if)# ipv6 ospf6 area 0
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth1
msk-frank-gw-03(config-if)# ipv6 ospf6 area 0
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth2
msk-frank-gw-03(config-if)# ipv6 ospf6 area 0
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# exit
msk-frank-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03#
```

msk-frank-gw-04 - PuTTY

```
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# router ospf6
msk-frank-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-frank-gw-04(config-ospf6)# exit
msk-frank-gw-04(config)# interface eth0
msk-frank-gw-04(config-if)# ipv6 osp6 area 0
% Unknown command: ipv6 osp6 area 0
msk-frank-gw-04(config-if)# ipv6 ospf6 area 0
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ipv6 ospf6 area 0
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04#
```

Захват из Standard input [TabouFrank-sw-01 Ethernet1 to msk-frank-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

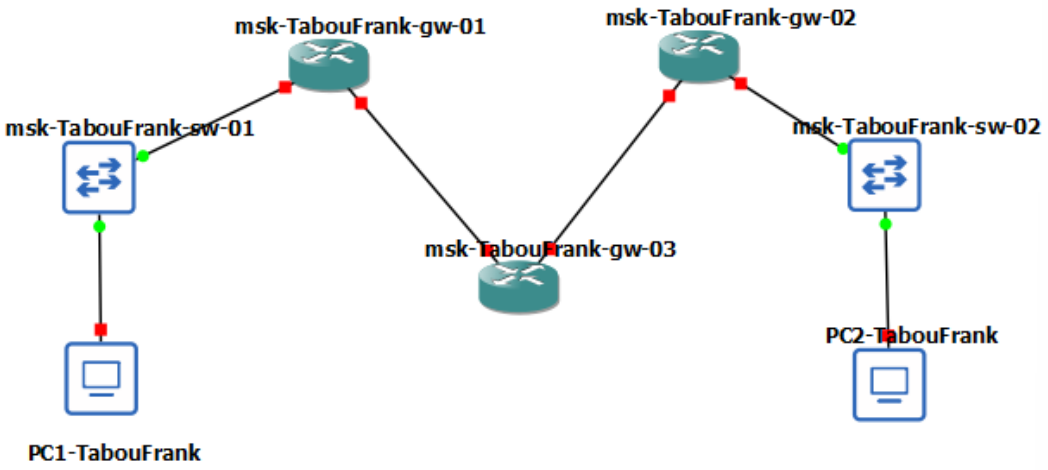
No.	Time	Source	Destination	Protocol	Length	Info
645	3198.808613	2001:10::a	2001:11::a	UDP	126	18480 → 18481 Len=64
646	3198.810608	2001:4::1	2001:10::a	ICMPv6	174	Time Exceeded (Hop limit exceeded in transit)
647	3198.811802	2001:10::a	2001:11::a	UDP	126	18480 → 18481 Len=64
648	3198.821050	2001:3::1	2001:10::a	ICMPv6	174	Time Exceeded (Hop limit exceeded in transit)
649	3198.821839	2001:10::a	2001:11::a	UDP	126	18480 → 18481 Len=64
650	3198.827434	2001:3::1	2001:10::a	ICMPv6	174	Time Exceeded (Hop limit exceeded in transit)
651	3198.828735	2001:10::a	2001:11::a	UDP	126	18480 → 18481 Len=64
652	3198.832468	2001:3::1	2001:10::a	ICMPv6	174	Time Exceeded (Hop limit exceeded in transit)
653	3198.833094	2001:10::a	2001:11::a	UDP	126	18480 → 18481 Len=64
654	3198.837054	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
655	3198.837282	2001:10::a	2001:11::a	UDP	126	18480 → 18481 Len=64
656	3198.841102	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
657	3198.841652	2001:10::a	2001:11::a	UDP	126	18480 → 18481 Len=64
658	3198.847822	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
659	3201.201688	fe80::e2e:91ff:fe2f…	ff02::9	RIPng	106	Command Response, Version 1
660	3203.498301	fe80::e2e:91ff:fe2f…	ff02::5	OSPF	90	Hello Packet
661	3203.643707	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
662	3213.499565	fe80::e2e:91ff:fe2f…	ff02::5	OSPF	90	Hello Packet
663	3213.644544	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
664	3220.436447	10.0.10.1	224.0.0.9	RIPv2	126	Response
665	3223.500795	fe80::e2e:91ff:fe2f…	ff02::5	OSPF	90	Hello Packet
666	3223.644952	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet

```
Frame 1: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0
    Section number: 1
  > Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Mar 13, 2026 20:10:55.995775000 RTZ 2 (зима)
    UTC Arrival Time: Mar 13, 2026 17:10:55.995775000 UTC
    Epoch Arrival Time: 1773421855.995775000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000000000 seconds]
    [Time delta from previous displayed frame: 0.000000000 seconds]
    [Time since reference or first frame: 0.000000000 seconds]
    Frame Number: 1
    Frame Length: 126 bytes (1008 bits)
    Capture Length: 126 bytes (1008 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
```

```
0000  01 00 5e 00 00 09 0c 2e  91 2f 00 00 08 00 45 c0   ··^····· ·/····E·
0010  00 70 ba 10 40 00 01 11  ca a2 0a 00 0a 01 e0 00   ·p··@··· ********
0020  00 09 02 08 02 08 00 5c  d7 06 02 02 00 00 00 02   ·******\ ********
0030  00 00 0a 00 01 00 ff ff  ff 00 00 00 00 00 00 00   ******** ********
0040  00 01 00 02 00 00 0a 00  02 00 ff ff ff 00 00 00   ******** ********
0050  00 00 00 00 00 03 00 02  00 00 0a 00 03 00 ff ff   ******** ********
0060  ff 00 00 00 00 00 00 00  00 02 00 02 00 00 0a 00   ******** ********
0070  04 00 ff ff ff 00 00 00  00 00 00 00 00 01         ******** ******
```

```
msk-frank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-01# show ipv6 ospf6 neighbor
Neighbor ID     Pri    DeadTime    State/IfState         Duration I/F[State]
4.4.4.4           1    00:00:37    Full/BDR              00:03:19 eth2[DR]
msk-frank-gw-01# show ipv6 ospf6 router
% Unknown command: show ipv6 ospf6 router
msk-frank-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64                    ::                          eth1 00:03:42
*N IA 2001:2::/64                    fe80::e12:bfff:fe9d:1       eth2 00:03:42
*N IA 2001:3::/64                    fe80::e12:bfff:fe9d:1       eth2 00:03:42
*N IA 2001:4::/64                    ::                          eth2 00:03:42
*N IA 2001:10::/64                   ::                          eth0 00:03:42
*N IA 2001:11::/64                   fe80::e12:bfff:fe9d:1       eth2 00:03:42
msk-frank-gw-01#
```

```
msk-frank-gw-01# show ipv6 ospf6 router
% Unknown command: show ipv6 ospf6 router
msk-frank-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64                    ::                          eth1 00:03:42
*N IA 2001:2::/64                    fe80::e12:bfff:fe9d:1       eth2 00:03:42
*N IA 2001:3::/64                    fe80::e12:bfff:fe9d:1       eth2 00:03:42
*N IA 2001:4::/64                    ::                          eth2 00:03:42
*N IA 2001:10::/64                   ::                          eth0 00:03:42
*N IA 2001:11::/64                   fe80::e12:bfff:fe9d:1       eth2 00:03:42
msk-frank-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64                    ::                          eth1 00:08:02
*N IA 2001:2::/64                    fe80::e12:bfff:fe9d:1       eth2 00:08:02
*N IA 2001:3::/64                    fe80::e12:bfff:fe9d:1       eth2 00:08:02
*N IA 2001:4::/64                    ::                          eth2 00:08:02
*N IA 2001:10::/64                   ::                          eth0 00:08:02
*N IA 2001:11::/64                   fe80::e12:bfff:fe9d:1       eth2 00:08:02
msk-frank-gw-01#
```



PC1-TabouFrank - PuTTY

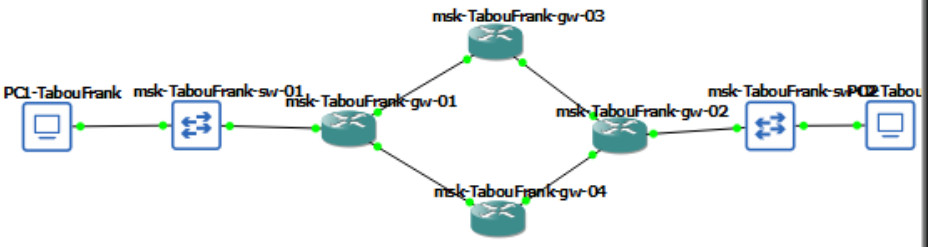
```
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 1000::a/64
PC1 : 1000::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 10000
RHOST:PORT          : 127.0.0.1:10001
MTU:                : 1500

VPCS>
```

PC2-TabouFrank - PuTTY

```
VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> sahow ipv6
Bad command: "sahow ipv6". Use ? for help.

VPCS> show ipv6

NAME              : VPCS[1]
LINK-LOCAL SCOPE  : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE      : 1002::a/64
DNS               :
ROUTER LINK-LAYER :
MAC               : 00:50:79:66:68:01
LPORT             : 10020
RHOST:PORT        : 127.0.0.1:10021
MTU:              : 1500

VPCS>
```

msk-TabouFrank-gw-01 - PuTTY

```
You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-TabouFrank-gw-0x
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-TabouFrank-gw-0x
[edit]
vyos@vyos# commit
sa[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
eDone
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] y

vyos@vyos:~$
```


msk-TabouFrank-gw-03
PC1-TabouFrank
msk-TabouFrank-sw-01
msk-TabouFrank-gw-01
msk-TabouFrank-gw-02
msk-TabouFrank-gw-04

8.3.3. Задание для самостоятельного выполнения

Заданы топология сети (рис. 8.6) и адреса сегментов сети (табл. 8.4).

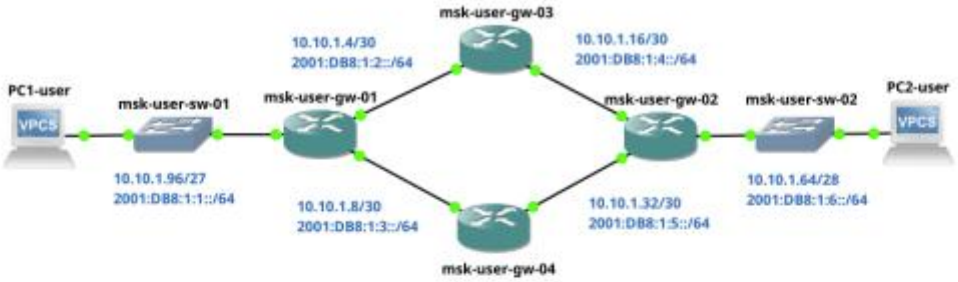


Рис. 8.6. Топология сети

Требуется:

- задокументировать моделируемую сеть: схемы L1, L3, заполнить таблицы адресации;
- настроить двойной стек IPv4 и IPv6, проверить соединения;
- настроить динамическую маршрутизацию сетей IPv4 и IPv6 для протоколов RIP, OSPF, проверить соединения и маршруты.

Таблица 8.4

Таблица адресов сетей

Устройства	Сеть IPv4	Сеть IPv6
PC1–gw-01	10.10.1.96/27	2001:db8:1:1::/64
PC2–gw-02	10.10.1.64/28	2001:db8:1:6::/64
gw-01–gw-03	10.10.1.4/30	2001:db8:1:2::/64
gw-02–gw-04	10.10.1.8/30	2001:db8:1:3::/64
gw-03–gw-02	10.10.1.16/30	2001:db8:1:4::/64
gw-04–gw-02	10.10.1.32/30	2001:db8:1:5::/64

```
[   20.318095] systemd[1]: Detected architecture x86-64.
[   20.341849] systemd[1]: Set hostname to <vyos>.
[   21.077323] systemd[1]: Listening on udev Control Socket.
[   21.087302] systemd[1]: Listening on udev Kernel Socket.
[   21.097306] systemd[1]: Created slice system-serial\x2dgetty.slice.
[   21.114478] systemd[1]: Created slice User and Session Slice.
[   21.141350] systemd[1]: Started Dispatch Password Requests to Console Directo
ry Watch.
[   21.154598] systemd[1]: Listening on Journal Socket.
[   21.361369] bridge: filtering via arp/ip/ip6tables is no longer available by
default. Update your scripts to load br_netfilter if you need this.
[   21.384653] Bridge firewalling registered
[   21.405751] mpls_gso: MPLS GSO support

Welcome to VyOS - vyos ttyS0

vyos login:
```

msk-TabouFrank-gw-01 - PuTTY

```
[   21.405751] mpls_gso: MPLS GSO support

Welcome to VyOS - vyos ttyS0

vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address dhcp
     hw-id 0c:b5:af:f4:00:00
 }
 ethernet eth1 {
     hw-id 0c:b5:af:f4:00:01
 }
 ethernet eth2 {
     hw-id 0c:b5:af:f4:00:02
 }
 loopback lo {
 }
[edit]
vyos@vyos#
```

msk-TabouFrank-gw-01 - PuTTY

```
You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address dhcp
     hw-id 0c:b5:af:f4:00:00
 }
 ethernet eth1 {
     hw-id 0c:b5:af:f4:00:01
 }
 ethernet eth2 {
     hw-id 0c:b5:af:f4:00:02
 }
 loopback lo {
 }
[edit]
vyos@vyos# set interfaces ethernet eth0 address 2011:db8:1:1::1/64
[edit]
vyos@vyos# set interfaces ethernet eth1 address 2001:db8:1:1::1/64
[edit]
vyos@vyos# set interfaces ethernet eth0 address 2001:db8:1:1::1/64
[edit]
vyos@vyos# set interfaces ethernet eth1 address 2001:db8:1:2::1/64
[edit]
vyos@vyos# set protocols ospf area 0 network 10.10.1.0/24
[edit]
vyos@vyos#
```

Выход

ходе выполнения лабораторной работы были изучены основные принципы маршрутизации в сетях IPv4 и IPv6, а также особенности настройки сетевого оборудования. Была построена модель сети, настроена адресация интерфейсов и проверена связь между узлами. Кроме того, были изучены принципы работы динамической маршрутизации и проведена проверка корректности передачи пакетов между сегментами сети. Полученные результаты показали, что правильная настройка маршрутизации обеспечивает эффективный обмен данными и стабильную работу сети.